Class 24

- A player i has K information sets

  ℓ-th information set has m_ℓ many actions

For a behavior strategy b_i, we have to specify

$$\sum_{\ell=1}^{K} (m_\ell - 1) \quad \text{numbers}$$

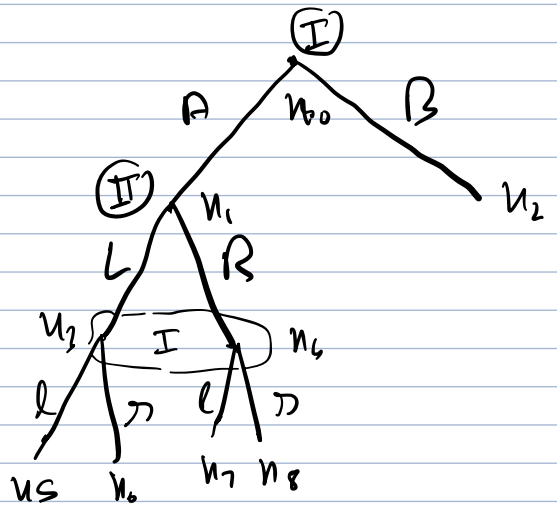
for a mixed strategy σ_i

$$\left(\prod_{\ell=1}^{K} m_\ell - 1 \right) \quad \text{numbers we have to specify}$$

Equivalence definition

A mixed strategy σ_i and a behavior strategy b_i of Player i in an extensive form game Γ, if for every mixed / behavior strategy vector σ_{-i} of the other players and for all vertex w of the game tree

$$\rho(w; (\sigma_i, \sigma_{-i})) = \rho(w; (b_i, \sigma_{-i}))$$

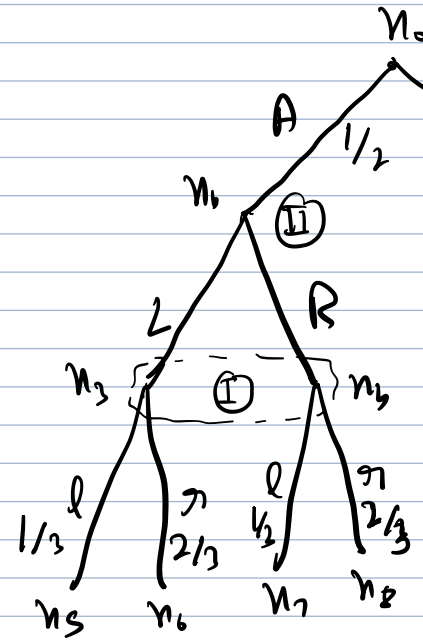


R-1 Suppose b_i (a behavior strategy) and σ_i (a mixed strategy) are equivalent then, for all mixed / behavior strategy vector σ_{-i} of the other players $\forall j$

$$U_j(\sigma_i, \sigma_{-i}) = U_j(b_i, \sigma_{-i}).$$

$$\Rightarrow U_j(\sigma_i, \sigma_i, \sigma_{-\{i,j\}}) = U_j(b_i, \sigma_j, \sigma_{-\{i,j\}})$$

R-2 Let $\sigma = (\sigma_i)_{i \in N}$ be a mixed strategy profile and $b = (b_i)_{i \in N}$ is a behavior strategy profile. Suppose b_i is equivalent to σ_i for all i. Then we have $U_i(\sigma) = U_i(b) \ \forall i$



$\sigma_1(Al) = \frac{1}{2} \times \frac{1}{3}$

$\sigma_1(Ar) = \frac{1}{2} \times \frac{2}{3}$

$\sigma_1(Bl) = 1/2 \times 1/3$

$\sigma_1(Br) = \frac{1}{2} \times \frac{2}{3}$

$\sigma_2(L) = 1$

$P(n_2; (\sigma_1, \sigma_2)) = P(n_2; (b_1, \sigma_2))$

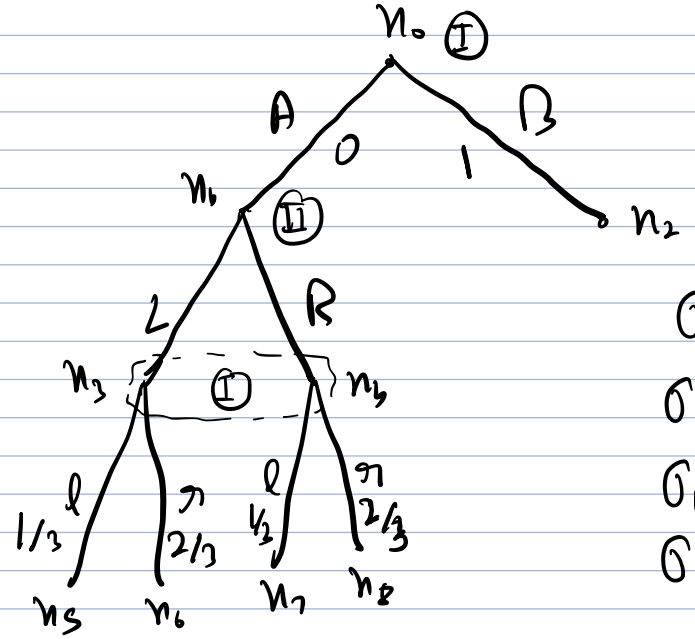
$$\rho(n_5;(\sigma_1,\sigma_2)) = \rho(n_5;(b_1,\sigma_2))$$

$$\rho(n_i,(\sigma_1,\sigma_2)) = \rho(n_i,(b_1,\sigma_2)) \quad \forall n_i \in L$$

- Given b_i, construct σ_i such that

$$\sigma_i(s_i) = \prod_{h_i \in H_i} b_i(s_i(h_i), h_i)$$

$$\sigma_i(s_i) = \prod_{h_i \in H_i} b_i(s_i(h_i), h_i)$$



$\sigma_1(Al) = 0$ 0

$\sigma_1(Ar) = 0$ 0

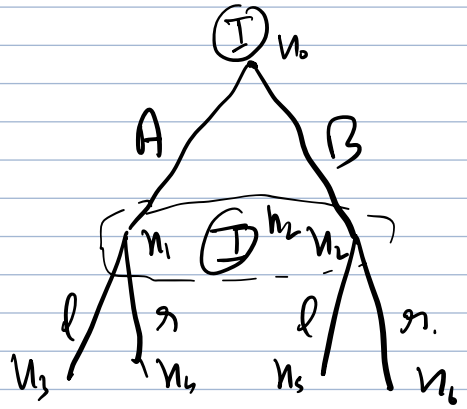
$\sigma_1(Bl) = 1/3$ α

$\sigma_1(Br) = 2/3$ $1-\alpha$

This equivalence is not unique in the sense that given a behaviour strategy, there can be multiple mixed strategies which are equivalent.

Theorem Given a behaviour strategy, we have an equivalent mixed strategy

The non-trivial question is that whether given a mixed strategy, an equivalent behavior strategy exists.



$\sigma_1(Al) = 1/2$

$\sigma_1(Bl) = 1/2$

$\sigma_1(Ar) = 0$

$\sigma_1(Br) = 0$

$b_1(A; u_0) = x$ $\quad b_1(l; h_2) = y$

$\rho(u_3; \sigma_1) = \frac{1}{2} = xy$ $\quad 0 = (1-x)y$

$\rho(u_4; \sigma_1) = 0 = x(1-y)$ $\quad 1/2 = (1-x)(1-y)$

if we dont have perfect recall, then there is a mixed strategy which does not have an equivalent behavior strategy.